

MRI Brain Tumor Classification Using Custom and Advanced CNN Models

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1. Introduction

Brain tumors represent a major global health concern due to their potential for rapid progression, severe neurological impact, and the complexity of their diagnosis and treatment. Early and accurate detection plays a crucial role in improving patient survival and treatment planning. Magnetic Resonance Imaging (MRI) is one of the most widely used diagnostic imaging techniques for brain tumors because it provides exceptional soft-tissue contrast, high-resolution anatomical detail, and does not involve ionizing radiation. MRI scans allow clinicians to visualize tumor boundaries, assess malignancy, and understand tumor behavior across multiple imaging modalities (T1, T2, FLAIR, etc.).

However, manual interpretation of MRI scans is time-consuming, subjective, and highly dependent on the radiologist's expertise. Subtle variations in tumor appearance, overlapping features among different tumor classes, and the presence of artifacts often make classification challenging. This complexity has motivated the use of artificial intelligence—particularly deep learning—to support radiologists in making faster, more accurate, and more consistent diagnostic decisions.

In recent years, Convolutional Neural Networks (CNNs) have emerged as the leading deep learning approach for medical image analysis. CNNs have the unique ability to automatically learn hierarchical image features such as edges, textures, shapes, and high-level structures, eliminating the need for manual feature extraction. Their powerful pattern-recognition capability has led to breakthrough performance in tasks like tumor detection, segmentation, and classification.

This project builds upon these advancements by designing and evaluating a custom CNN model specifically tailored for brain tumor classification using MRI images. The focus is on classifying images into four clinically relevant categories:

Glioma – tumors originating from glial cells, often aggressive

Meningioma – tumors arising from the meninges, typically benign but space-occupying

Pituitary tumor – tumors located in or near the pituitary gland

Non-tumor – healthy brain scans or scans with no visible tumor

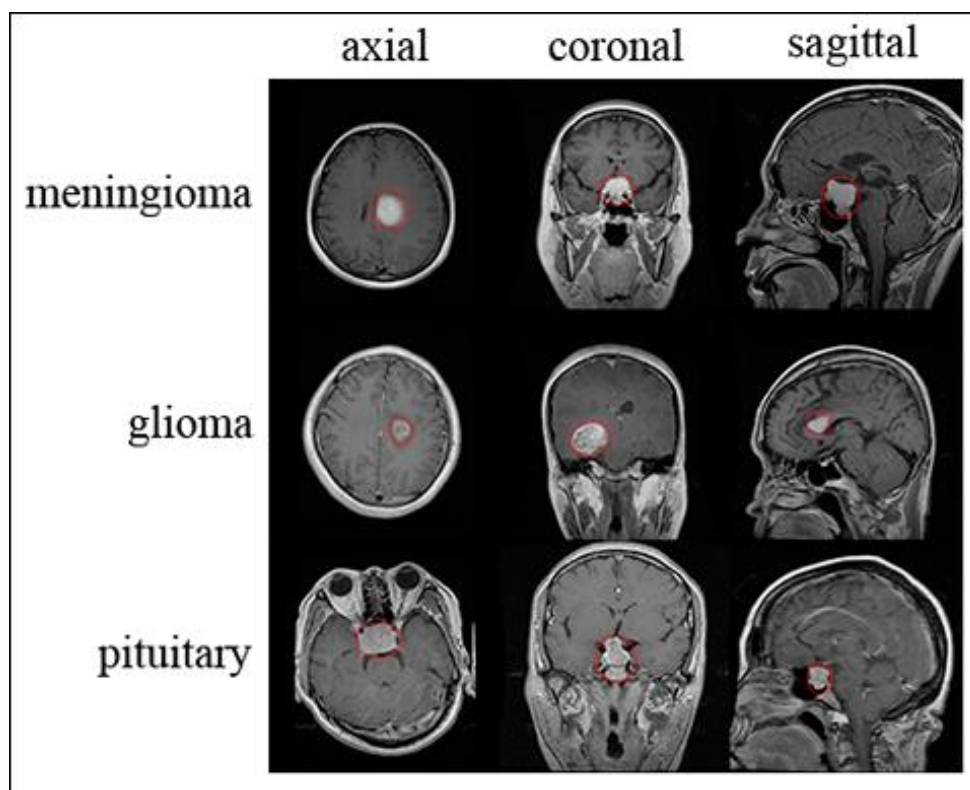


Figure 1 Different types of brain tumours used for classification

The aim of this work is to establish a strong baseline deep learning model that can effectively learn visual patterns associated with each category. By systematically preprocessing the dataset, constructing a custom CNN architecture, training the model from scratch, and thoroughly evaluating its performance, the project demonstrates how deep learning can contribute to clinical decision-support systems. Ultimately, the model developed in this study

has the potential to improve diagnostic accuracy, reduce manual workload, and serve as a foundation for future enhancements such as transfer learning, multimodal MRI analysis, or integration into real-world radiology workflows.

2. Objectives of the Project

The primary objective of this project is to develop and evaluate deep learning models capable of accurately classifying brain tumors using MRI image data. Brain tumor identification is a complex task due to variations in tumor appearance, size, and location, making automated techniques especially valuable. This project aims to design a robust baseline system using a custom Convolutional Neural Network (CNN) architecture tailored for four-class classification involving Glioma, Meningioma, Pituitary Tumor, and Non-tumor categories. A major objective is to implement effective preprocessing strategies to standardize MRI images, including resizing them to a uniform dimension, normalizing pixel intensities for stable learning, and adjusting brightness or contrast to reduce variations across samples. These steps ensure that the CNN model receives consistent and high-quality input data. Another key goal is to construct the CNN architecture itself, incorporating convolutional and pooling layers that progressively extract meaningful features from the MRI scans, followed by fully connected layers that perform final classification. The project further aims to train, validate, and test the model using an appropriate dataset split while monitoring metrics such as accuracy, loss, precision, recall, and F1-score to track performance throughout training. This evaluation process helps identify overfitting, underfitting, or class imbalance issues. Finally, the project seeks to analyze the model's predictive performance across different tumor types, investigate misclassification patterns, and assess whether the proposed CNN demonstrates sufficient generalization and consistency to support clinical applicability. This comprehensive evaluation helps determine

the model’s potential for assisting radiologists and improving diagnostic decision-making in real-world healthcare settings.

3. Study Design and Methodology

3.1 Data Preprocessing

The MRI dataset used in this project consists of four major categories of brain tumor images, making preprocessing an essential step for ensuring high-quality input to the deep learning model. To achieve uniformity, all MRI scans were resized to a fixed input dimension, such as

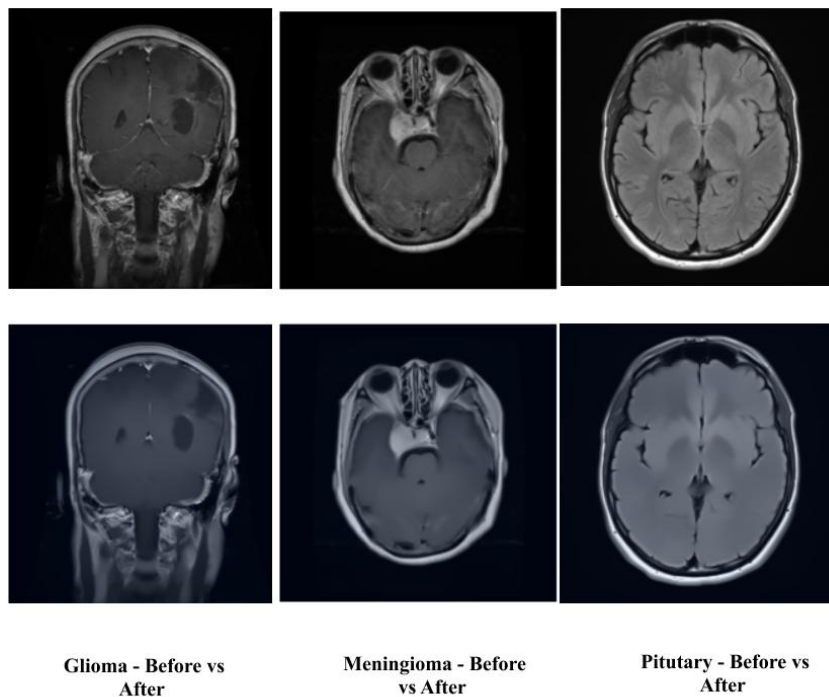


Figure 2 Before and after preprocessing - smoothing of the image

150×150 pixels, allowing the CNN to process images consistently regardless of their original resolution. Pixel intensity normalization was applied by scaling values to the 0–1 range, which helps stabilize the training process and improves the performance of gradient-based optimization. In cases where MRI scans exhibited variations in brightness or minor artifacts,

intensity correction techniques were used to enhance contrast and reduce noise, ensuring that meaningful tumor features were preserved.

The dataset was then divided into training, validation, and testing subsets using stratified splitting. This ensured that each tumor category—Glioma, Meningioma, Pituitary Tumor, and Non-tumor—was proportionally represented across all subsets, preventing class imbalance during training and evaluation. Additionally, data visualization techniques were employed to inspect a selection of images from each class. This step provided valuable insights into variations in tumor shapes, textures, and intensities, helping validate the preprocessing pipeline and guiding model design decisions.

3.2 Building the Custom CNN Model

A custom Convolutional Neural Network (CNN) architecture was designed as the baseline model for brain tumor classification. The model consists of sequential convolutional layers with ReLU activation, each followed by max-pooling to progressively extract and compress important spatial features from the MRI images. After the feature extraction stages, the output is flattened and passed through a fully connected dense layer to learn higher-level representations. Finally, a softmax output layer is used to classify images into four categories: Glioma, Meningioma, Pituitary Tumor, and Non-tumor. This structure provides an effective foundation for multiclass medical image classification.

3.3 Training the Model

The custom CNN model was trained from scratch using the fully preprocessed training dataset, ensuring that the network learned feature representations directly from the MRI images. The training process employed the Adam optimizer, chosen for its efficiency and adaptive learning rate capabilities, which help accelerate convergence during deep learning tasks. Categorical cross-entropy was used as the loss function, as it is well-suited for multiclass classification problems involving the four tumor categories. Throughout the training phase, key metrics such as accuracy and loss were systematically recorded and analyzed for each epoch. Tracking these metrics enabled close monitoring of the model's learning behavior, helping identify issues such as slow convergence, underfitting, or overfitting.

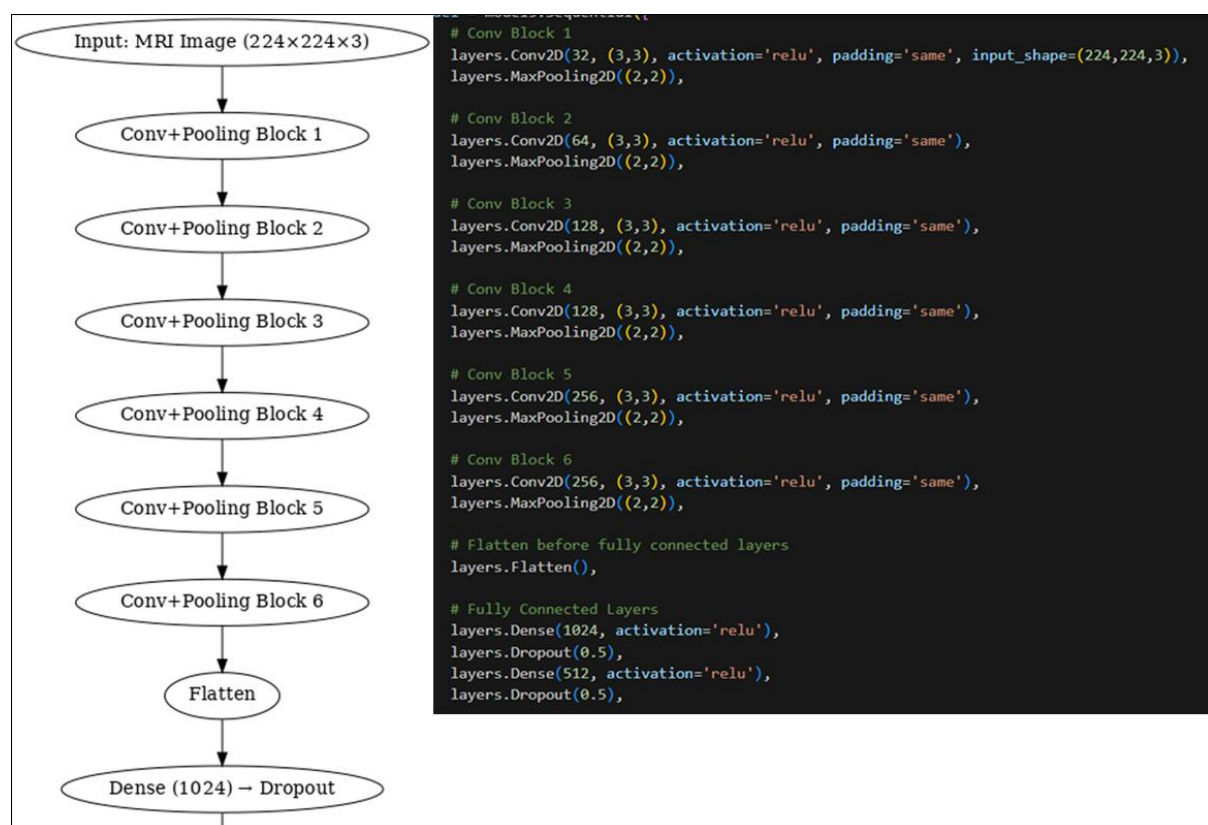


Figure 3 Differnt CNN layers applied

To enhance the model's generalization ability, data augmentation techniques—such as random rotations, zooming, flipping, and shifting—were optionally applied to artificially increase dataset variability. This approach exposes the model to different orientations and variations of the same image, reducing overfitting and improving robustness. Validation metrics were evaluated alongside training performance to ensure consistent improvement and maintain balance between learning and generalization. Overall, the structured training pipeline allowed the model to progressively refine its feature-learning capability and improve its classification accuracy across the four brain tumor categories.

3.4 Model Evaluation

The validation dataset played a crucial role during the training phase by providing an unbiased evaluation of the model's performance at each stage. While the training data guided the model's learning, the validation data offered insights into how well the model generalized to unseen samples. Key performance metrics—including accuracy, precision, recall, and F1-score—were computed during validation to obtain a comprehensive understanding of the classifier's behaviour. Accuracy provided an overall measure of correct predictions, while precision and recall highlighted the model's ability to correctly identify tumor classes without producing excessive false positives or false negatives. The F1-score, being the harmonic mean of precision and recall, offered a balanced measure of performance, especially useful when class distributions varied. Monitoring these metrics throughout training helped identify strengths, detect early signs of overfitting, and pinpoint weaknesses in differentiating specific tumor types. This continuous feedback guided model tuning and ensured stable, reliable performance.

3.5 Testing the Model

After completing the training process, the final saved model was evaluated on a completely separate test dataset to measure its real-world performance and generalization capability. This independent evaluation provided an unbiased understanding of how well the model could classify new MRI images. The analysis focused first on the overall test accuracy

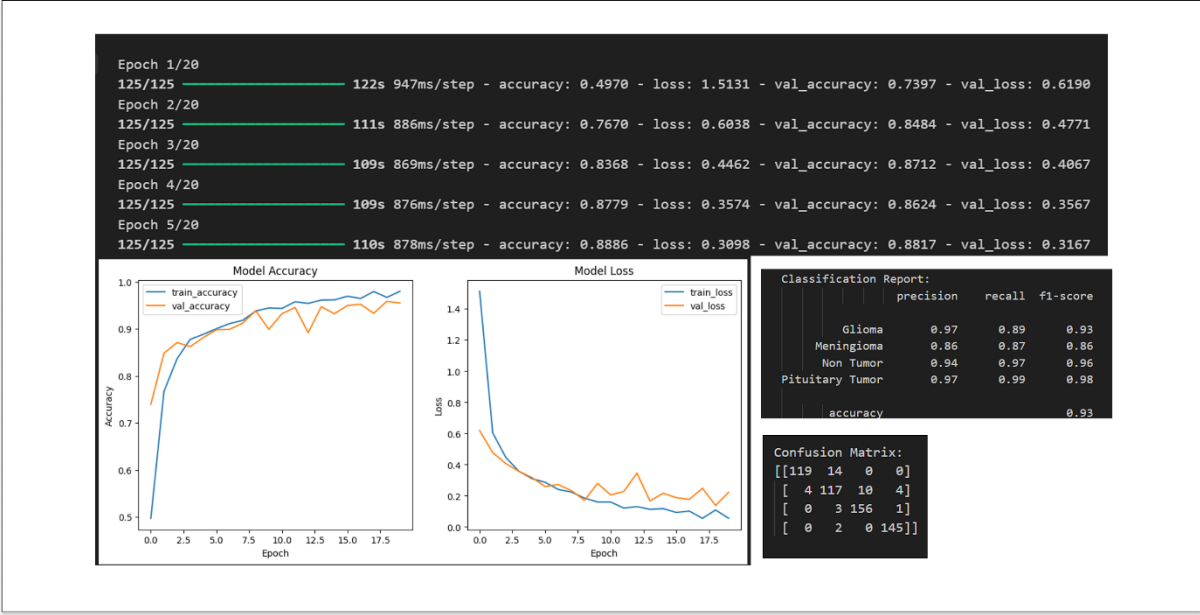


Figure 4 Result and evaluation of the model

to determine the model’s effectiveness across all four tumor categories. In addition, class-wise precision, recall, and F1-score were calculated to examine how the model performed on each individual tumor type, highlighting strengths and potential weaknesses. A comparison between validation and test metrics was also conducted to ensure consistency and confirm that the model did not overfit during training.

Misclassified cases were closely inspected to explore possible reasons behind errors, such as overlapping tumor features, low-quality images, or ambiguous boundaries. Finally, the

results were reviewed to assess the model's potential clinical applicability, emphasizing the stability, reliability, and limitations of its performance in a real diagnostic setting.

4. Conclusion

In conclusion, this project successfully developed a custom CNN model capable of classifying brain tumors from MRI images with promising accuracy. Through systematic preprocessing, thoughtful model design, and rigorous evaluation, a strong baseline was established for automated tumor detection. The results highlight the significant potential of deep learning in supporting clinical decision-making and improving diagnostic workflows. However, the study also reveals opportunities for further enhancement, including expanding the dataset, incorporating transfer learning, and exploring deeper or hybrid architectures. Overall, the project provides a solid foundation for future advancements in medical image analysis and AI-driven healthcare applications.